Distance	States s in S processing order Q	Hash rank h(S)
0	[1234]	0
1	[1243, 1432]	1, 5
2	[4132, 4213]	19, 20
3	[2413, 3142]	10, 13
4	[2314, 3124]	8, 12
5	[2341, 3421]	9, 17
6	[4321]	23

S	h(S)	Level of d[S]	π[S]	move[S]	States N of Q for next level
1234	0	0	Nil	Nil	[1243, 1432]
1243	1	1	1234	L	
1432	5	1	1234	U	[4132, 4213]
4132	19	2	1432	L	
4213	20	2	1243	U	[2413, 3142]
2413	10	3	4213	R	
3142	13	3	4132	D	[2314, 3124]
2314	8	4	2413	D	

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3124	12	4	3142	R	[2341, 342]
2341	9	5	2314	L	
3421	19	5	3124	U	[4321]
4321	23	6	2341	U	

God's number for the puzzle

God's Number:

$$G(S) = \max_{S\text{-reachable states}} d(S)$$

$$= \max\{0, 1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 6\}$$

(a multiset)

$$= \underline{\underline{6}}$$

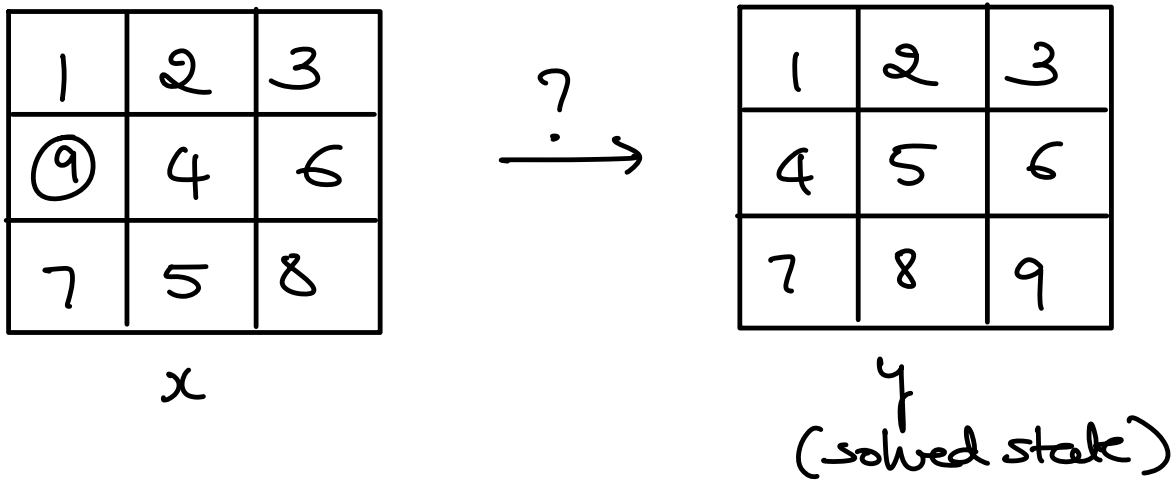
$n \times n$ Board	Reachable States	God's Number	n^3
2×2	$\frac{4!}{2} = 12$	6	$2^3 = 8$
3×3	$9!/2 = 181440$	31	$3^3 = 27$
4×4	$16!/2 = 10,461,394,944,000$	80	$4^3 = 64$
5×5	$25!/2 = 7.76 \times 10^{24}$	Unknown $140 \leq G \leq 210$	$5^3 = 125$
$n \times n$			

$G(n) = n^3 + O(n^{2.75})$ for an n-puzzle

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God's number for Rubik's cube (3x3) is 20.

3x3 Puzzle: (Algorithm for solving the 3x3 puzzle)



Whether y is attainable from x ?

To check: $I(x) = I(y)$?

$$I(x) = I(123946758)$$

$$= \text{sgn}(x) \cdot \chi(x)$$

$$= (-1)^{9} \cdot (-1)^{2+1}$$

$$= (-1) \cdot (-1)$$

$$= \underline{1}$$

Inversions(x): 123946758

0 0 0 5 0 1 1 0 0 ← Factorial digits

Sum = 9

$\chi(x) = (-1)^{\text{row + column (where 9 (blank) is placed in } x)}$

$= (-1)^{3+2} = -1.$

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$$I(y) = I(123456789)$$

$$= \text{sgn}(y) \cdot X(y)$$

$$= (-1)^{0} \cdot (-1)^{3+3}$$

$$= 1 \cdot 1$$

$$= \underline{\underline{1}}$$

Inversions (123456789) :

0 00000000 (Factorial Digits)

row = 3, column = 3.

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Hence $I(x) = I(y) = 1$.

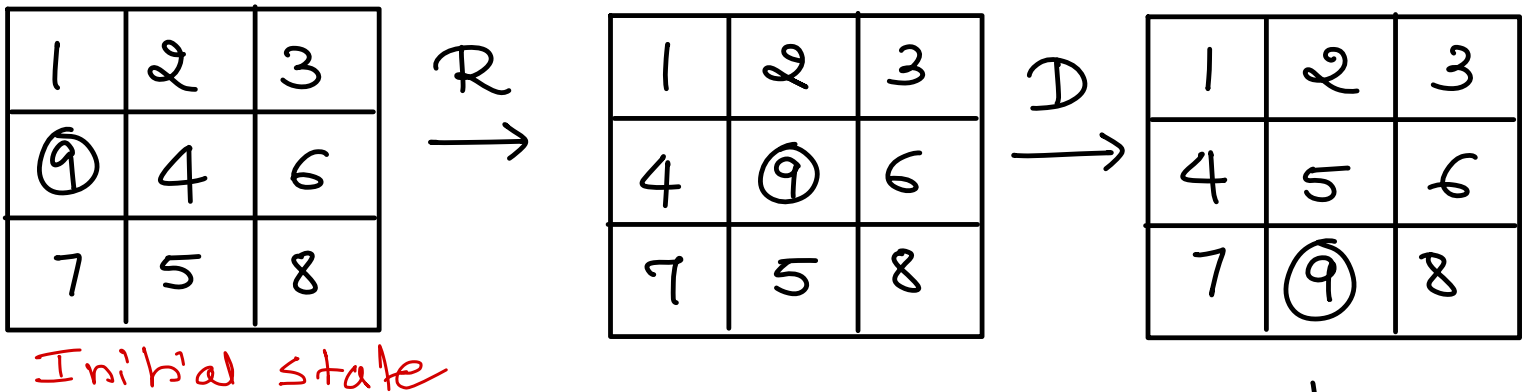
Therefore puzzle is solvable

A^* Algorithm :

$$f(n) = g(n) + h(n)$$

- $f(n)$: Estimated total cost of the path through n (current node)
- $g(n)$: Actual cost from start node to n (current node) (g-score) ↓ No of nodes traversed from start state to current node.
- $h(n)$: Heuristic estimate of cost from n to the goal (h-score) ↓ How far the goal node is.

optimal solution is



Initial state

Goal state

We will prove this using A* Algorithm

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